



< Previous







Next >

6.1.1 Whose problem is it anyway?

 Bookmark this page

Pursue a verified certificate



Verified Certificate

THE UNIVERSITY of TEXAS SYSTEM

This is to certify that

Sandipan Dey

successfully completed and received a passing grade in

UT.ALA:Advanced Linear Algebra: Foundations to Frontiers

a course of study offered by UTAustinX, an online learning initiative of University of Texas System.

 **Maurilio Pugliesi**
Professor
UniversityX

 **Justine Doe, Ph.D.**
Professor
UniversityX

 **Verified Certificate**

Valid Certificate ID

Issued Month / Year

12345678901234567890

→ Unlimited access to all course materials

→ Shareable [verified certificate](#)

→ AI learning assistant and translations

Upgrade for \$99

Reading assignment

1/1 point (graded)



Read Unit 6.1.1 of the notes. [\[LINK\]](#)

☒ done



Submit

✓ Correct (1/1 point)

Discussion

Hide Discussion

Topic: Week 06 / 06.1.1

Add a Post

Show all posts

by recent activity

There are no posts in this topic yet.

✕

Homework 6.1.1.2

1/1 point (graded)

You use some linear system solver and it gives the wrong answer. In other words, you solve $Ax = b$ on a computer, computing $\hat{x}$, and somehow you determine that

$$\|x - \hat{x}\|$$

is large. Which of the following is a possible cause (identify all):

Mark all.

- ☒ There is a bug in the code. In other words, the algorithm that is used is sound (gives the right answer in exact arithmetic) but its implementation has an error in it.
- ☒ The linear system is ill-conditioned. A small relative error in the right-hand side can amplify into a large relative error in the solution.
- ☒ The algorithm you used accumulates a significant roundoff error.
- ☒ All is well: $\|\hat{x} - x\|$ is large but the relative error $\|\hat{x} - x\|/\|x\|$ is small.



Solution:

For a complete solution, see [the notes](#).

Submit

< Previous

Next >

© All Rights Reserved



edX

- [About](#)
- [Affiliates](#)
- [edX for Business](#)
- [Open edX](#)
- [Careers](#)
- [News](#)

Legal

- [Terms of Service & Honor Code](#)
- [Privacy Policy](#)
- [Accessibility Policy](#)
- [Trademark Policy](#)
- [Sitemap](#)
- [Cookie Policy](#)
- [Your Privacy Choices](#)

Connect

- [Idea Hub](#)
- [Contact Us](#)
- [Help Center](#)
- [Security](#)
- [Media Kit](#)

